

Ex.no:6B	DATA VISUALIZATION USING POWER BI – (RETAIL_SALES_RAW)

Aim:

To Load the data(Retail_sales) into PowerBI and transform the data and visualize it to get insights

Algorithm:

ALGORITHM 1: IMPORTING DATA INTO POWER BI

Steps:

1. Open Power BI Desktop
2. Select:Home → Get Data
3. Choose Excel or CSV
4. Browse dataset
5. Click Load
6. Verify imported data

ALGORITHM 2: DATA TRANSFORMATION USING POWER QUERY

Steps:

1. Select **Transform Data**
2. Open Power Query Editor
3. Remove duplicates
4. Handle null values (replace blanks/nulls in columns like CustomerName and Discount)
5. Rename columns & format text casing (e.g., Capitalize Region)
6. Apply changes & Close

ALGORITHM 3: ADVANCED TRANSFORMATIONS & DATE PARSING

Steps:

1. Select OrderDate column and ensure data type is set to **Text**
2. Go to **Add Column** → **Custom Column**
3. Name column: ParsedDate
4. Apply Power Query M Formula:

Code snippet:

```
a. if Text.StartsWith([OrderDate], "2025-") then
    i. Date.FromText([OrderDate])
b. else if Text.Contains([OrderDate], "/") then
    i. Date.FromText([OrderDate], [Format = "dd/MM/yyyy", Culture =
        "en-GB"])
c. else
    i. Date.FromText([OrderDate], [Format = "MM-dd-yyyy", Culture =
        "en-US"])
```

5. Handle negative quantities (Absolute Value or filter rows)
6. Click **Close & Apply** to load the clean dataset into the model

ALGORITHM 4: CREATING CALCULATED COLUMN (REVENUE)**Steps:**

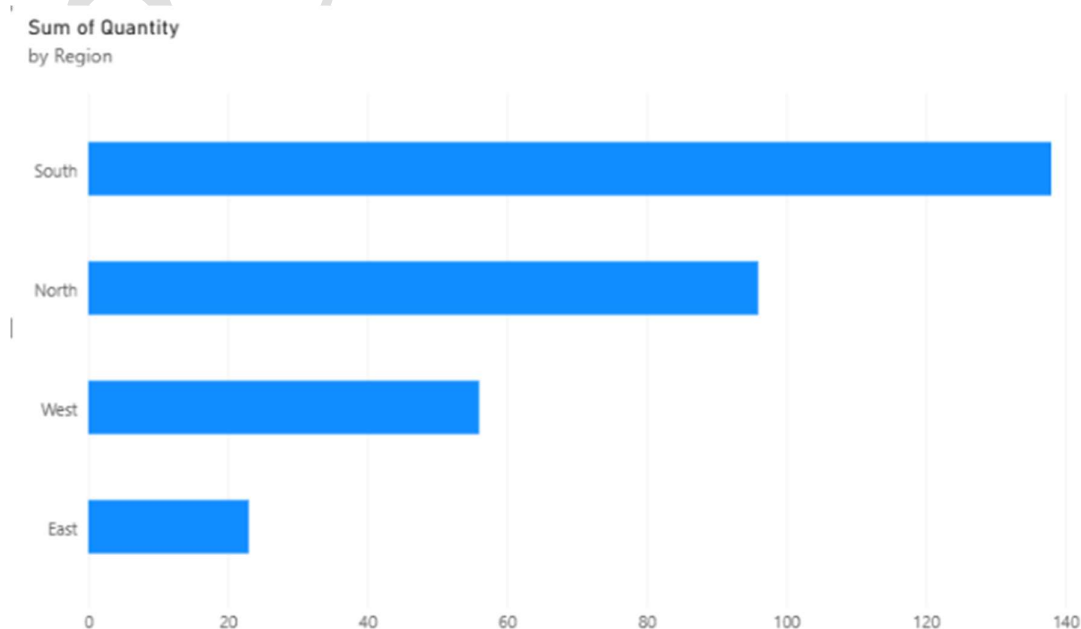
1. Navigate to **Report View** or **Table View**
2. Right-click dataset table and select **New Column**
3. Enter Revenue formula:

$$\text{Revenue} = [\text{Quantity}] \times [\text{UnitPrice}] \times (1 - (\text{if } [\text{Discount}] = \text{null then } 0 \text{ else } [\text{Discount}]))$$

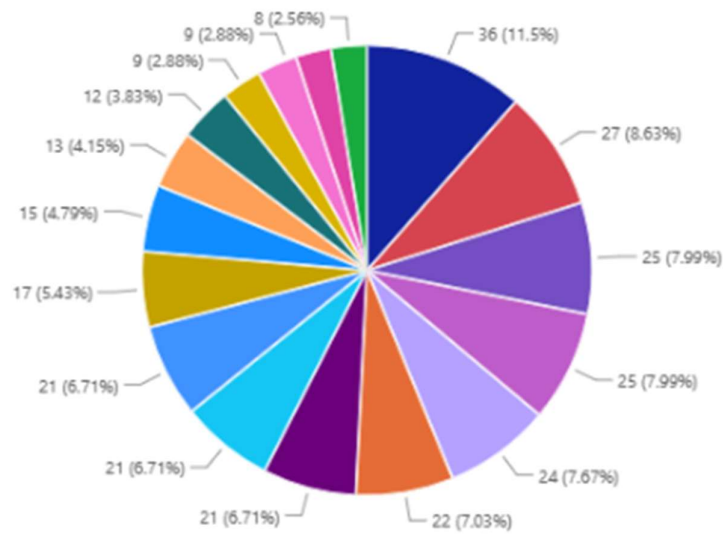
4. Press **Enter** to validate and compute values

ALGORITHM 5: DASHBOARD VISUALIZATION & REPORTING**Steps:**

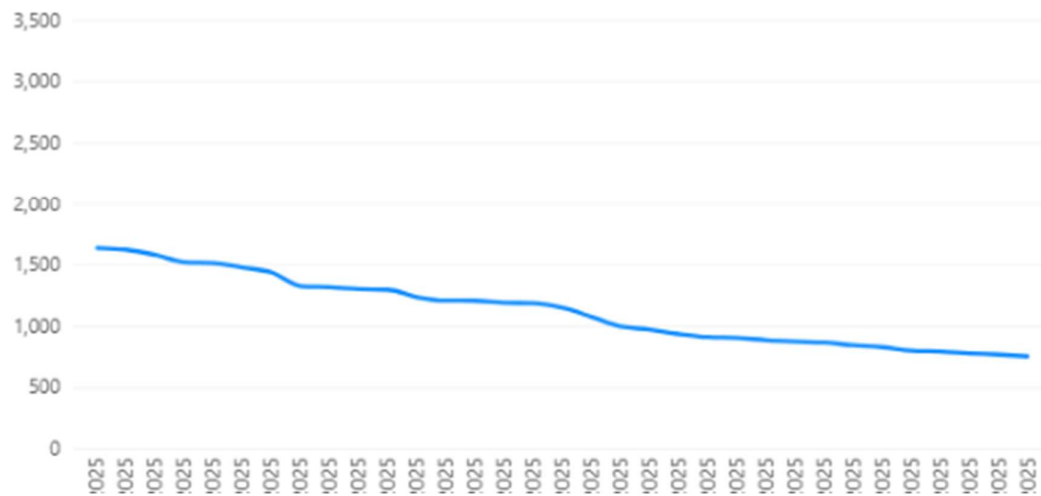
1. Select **Line Chart** visualization and map ParsedDate (X-axis) and Revenue (Y-axis)
2. Select **Bar / Column Chart** and map Region (X-axis) and Revenue / Quantity (Y-axis)
3. Select **Donut Chart** to break down quantities by Product
4. Add **Slicer** visuals for Region and Date to enable interactive filtering
5. Save the Power BI file (.pbix)

Output:

Sum of Quantity
by Product



Sum of Revenue
by ParsedDate

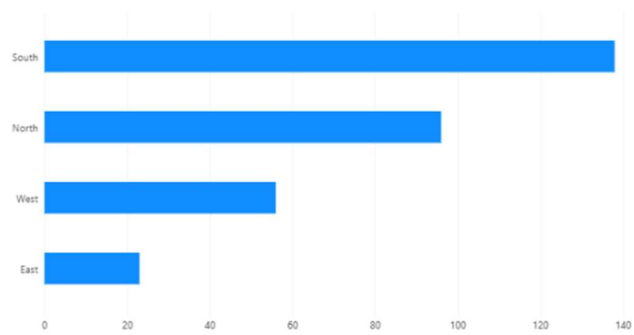


Sum of Revenue
74.31K

Region

- ☐ East
- ☐ North
- ☐ South
- ☐ West

Sum of Quantity
by Region

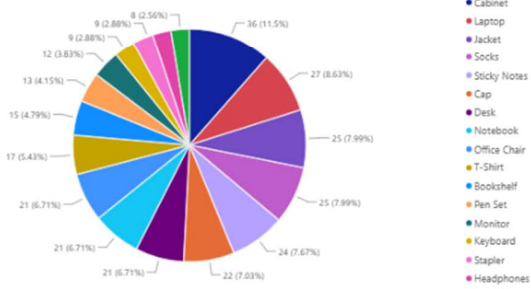


Sum of Revenue
74.31K

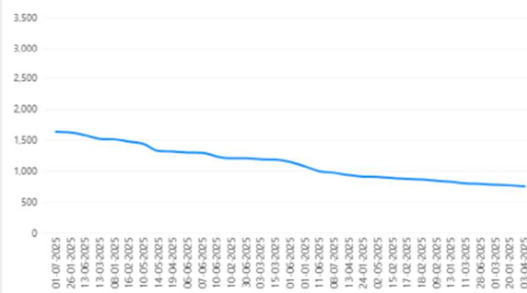
Region

- ☐ East
- ☐ North
- ☐ South
- ☐ West

Sum of Quantity
by Product



Sum of Revenue
by ParsedDate



Result: